COMP233 – Discrete Mathematics

Chapter 4: Number Theory and Methods of Proof

Lecture 4.4: Quotient-Remainder Theorem

Representing integers using quotient and remainder and applications to proofs and absolute value.

1 Motivation and Division

Exact Division

If d divides n ($d \mid n$), then there exists an integer k such that $n = dk$ and the remainder is 0.

When division is not exact

For any integer n and positive integer d, we can still write

$$n = dq + r,$$

where r is the amount left over (the **remainder**) with $0 \le r < d$.

2 Quotient-Remainder Theorem

Theorem 4.4.1 (Quotient-Remainder Theorem)

For every integer n and every positive integer d, there exist unique integers q and r such that

$$n = dq + r, \qquad 0 \le r < d.$$

q : Quotient

Number of complete groups of size d in n.

r : Remainder

Amount left over, with $0 \le r < d$.

Uniqueness

The integers q and r are unique. If $n = dq + r = dq' + r'$ with $0 \le r, r' < d$, then $q = q'$ and $r = r'$.

3 Computing div and mod

The quotient and remainder are defined by

$$n \textbf{ div } d = q, \qquad n \textbf{ mod } d = r$$

where $n = dq + r$ and $0 \le r < d$.

Example 1: 32 div 9 and 32 mod 9

$$32 = 9(3) + 5$$
$$32 \text{ div } 9 = 3, \qquad 32 \bmod 9 = 5.$$

Examples with negative integers

n	d	$q = n \text{ div } d$	$r = n \bmod d$	Check
54	4	13	2	$54 = 4(13) + 2$
-54	4	-14	2	$-54 = 4(-14) + 2$
54	70	0	54	$54 = 70(0) + 54$
-17	5	-4	3	$-17 = 5(-4) + 3$

4 Mathematical mod vs. Programming %

Mathematical mod (Epp)

- Remainder satisfies $0 \le r < d$.
- Always nonnegative.
- Defined by $n = dq + r$.

Programming %

- In C++ and Java, integer division truncates toward zero.
- % can be negative for a negative dividend.
- Behavior may differ from mathematical **mod**.

Expression	Mathematical result	C++/Java result
$-11 \bmod 4$	1	-3 $(-11 \,\%\, 4)$
$-17 \bmod 5$	3	-2 $(-17 \,\%\, 5)$
$6 \bmod 4$	2	2 $(6 \,\%\, 4)$

Key Point

- In this course, n **mod** d always denotes the **mathematical remainder** with $0 \le r < d$.
- Be careful when using % in programming languages.

5 Representing Integers by Remainders

Modulo 2: Even and Odd Integers

Every integer is either even or odd:

$n = 2q$ (even) or $n = 2q + 1$ (odd)

for some integer q.

These two cases are exhaustive and mutually exclusive.

Example

- $28 = 2(14) \rightarrow$ even
- $17 = 2(8) + 1 \rightarrow$ odd
- $-7 = 2(-4) + 1 \rightarrow$ odd

Modulo 4: Four Possible Remainders

Every integer has exactly one remainder when divided by 4. Thus every integer is one of the following forms:

Remainder 0	Remainder 1	Remainder 2	Remainder 3
$4q$	$4q + 1$	$4q + 2$	$4q + 3$

These four cases are exhaustive and mutually exclusive.

Examples

- $20 = 4(5)$ (remainder 0)
- $13 = 4(3) + 1$ (remainder 1)
- $-6 = 4(-2) + 2$ (remainder 2)
- $11 = 4(2) + 3$ (remainder 3)

6. Applications

Days of the Week (mod 7)

If today is day n, the day after k days is given by $(n + k) \bmod 7$.

Example

If today is Tuesday ($n = 2$) and $k = 25$, then $(2 + 25) \bmod 7 = 27 \bmod 7 = 6$, which is Saturday.

Modular Arithmetic Example

Find the remainder when 7^{50} is divided by 5.

Solution

$7 \bmod 5 = 2$. Since $2^4 = 16 \equiv 1 \pmod 5$,
$7^{50} \equiv 2^{50} = (2^4)^{12} \cdot 2^2 \equiv 1^{12} \cdot 4 \equiv 4 \pmod 5$.
So the remainder is 4.

Other Applications

- Cyclic structures (e.g., clocks, calendars, hashing).
- Cryptography and coding.
- Checking divisibility using remainders.

Example: Checking divisibility

n is divisible by $d \iff n \textbf{ mod } d = 0$.

- $35 \bmod 5 = 0 \rightarrow 5 \mid 35$
- $37 \bmod 5 = 2 \rightarrow 5 \nmid 37$

Example: Find 4m mod 11 given m mod 11 = 6

Since $m \equiv 6 \pmod{11}$,
$4m \equiv 4 \cdot 6 = 24 \equiv 2 \pmod{11}$.
So $4m \bmod 11 = 2$.

7 Proof by Division into Cases

Method

1. **Define cases**
   List all possible cases (e.g., using remainders).
2. **Prove each case**
   Prove the statement for each case separately.
3. **Conclude**
   Since the cases are exhaustive, the statement is true for all integers.

Example: Square of an odd integer

Prove: $\forall n \in \mathbb{Z}$, if n is odd, then $n^2 = 8m + 1$ for some integer m.

Proof: An odd integer is either $4q + 1$ or $4q + 3$.

Case 1: $n = 4q + 1$

$$n^2 = (4q + 1)^2 = 16q^2 + 8q + 1 = 8(2q^2 + q) + 1.$$

Case 2: $n = 4q + 3$

$$n^2 = (4q + 3)^2 = 16q^2 + 24q + 9 = 8(2q^2 + 3q + 1) + 1.$$

In both cases, the expression in parentheses is an integer. Hence $n^2 = 8m + 1$.

8. Absolute Value

Definition

$$|x| = \begin{cases} x, & x \ge 0 \\ -x, & x < 0 \end{cases}$$

- $|x|$ is the distance from 0.
- $|x| \ge 0$ and $|x| = 0$ iff $x = 0$.

Basic Properties

1. $|x| \ge 0$, and $|x| = 0$ iff $x = 0$.
2. $-|x| \le x \le |x|$.
3. $|-x| = |x|$.
4. If $|x| = |y|$, then $x = y$ or $x = -y$.
5. $|xy| = |x||y|$.

Triangle Inequality

$$|x + y| \le |x| + |y|$$

Proof (by cases on $x + y$):

- If $x + y \ge 0$, then $|x + y| = x + y \le |x| + |y|$.
- If $x + y < 0$, then $|x + y| = -(x + y) = (-x) + (-y) \le |x| + |y|$.

Example

Let $x = -3$, $y = 5$.

- $|x + y| = |2| = 2$,
- $|x| + |y| = 3 + 5 = 8$.
- $2 \le 8$.

Note

Equality does not always hold. For example, $x = 1$, $y = -1$:
$|x + y| = 0$, $|x| + |y| = 2$.

9. Common Misconceptions

Misconception	Correct Understanding
The remainder may be negative.	In mathematical **mod**, $0 \le r < d$.
div is always truncation toward zero.	n **div** $d = q$ where $n = dq + r$ and $0 \le r < d$.
Checking a few numbers proves a universal claim.	Use the quotient-remainder theorem to cover all integers.
Cases can be arbitrary.	Cases must be exhaustive and mutually exclusive.
$\lvert -x \rvert = -x$ for all x.	$\lvert -x \rvert = \lvert x \rvert$, but only equals $-x$ when $x < 0$.

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